

Power BI Project on Supply Chain Analytics

Week 3 – Advanced DAX & Analytical Modelling

Final Project Documentation

Power BI Advanced Analytics

Week 3

Prepared By: Khushi Sahni

1. Overview

This report is a documentation of the entire analytical assignment done by the Team B during Week 3 of the Supply Chain Analytics programme. This week was centered on the further development of DAX - the creation of time intelligence, rolling averages, risk scoring, segmentation, drillthrough pages, bookmarks and dynamic measures, based on the current Power BI data model built on the DataCo Supply Chain Dataset.

There were modules that were owned by each member of the team. All modules have been combined in this document as one structured reference and the Risk Scoring section is the formal deliverable of the Risk Methodology.

Team Member	Module
Dhanlaxmi	Time Intelligence – YTD, QTD, YoY & DAX Documentation
Sairam	Rolling Window Analytics – 30-day & 90-day Averages
Karishma	Risk Scoring Model – Delay, Profit & Discount Factors
Arun Dande	Segmentation Logic – Customer Tiering & Product Segments
Shreeram Prajapati	Bookmarks & Dynamic Measures
Aditya	Drillthrough Pages
Khushi Sahni	Risk Methodology & This Final Documentation

2. Time Intelligence – YTD, QTD & YoY

Contributor: Dhanlaxmi

Time intelligence measures have been constructed to assess business performance on a year, quarter and period basis. Everything relies on an unending Date Table, which is an essential prerequisite with regard to inbuilt time intelligent functions in Power BI.

Date Table

A calendar table was created spanning the full range of order dates in the dataset:

```
DateTable =  
CALENDAR(  
    MIN('DataCoSupplyChainDataset'[order date (DateOrders)]),  
    MAX('DataCoSupplyChainDataset'[order date (DateOrders)])  
)
```

Supporting columns added to the Date Table:

- Year = YEAR(DateTable[Date])
- Quarter = "Q" & FORMAT(DateTable[Date], "Q")
- Month Number = MONTH(DateTable[Date])

- Month Name = FORMAT(DateTable[Date], "MMMM") - sorted by Month Number for correct chart ordering

Base Measures

```
Total Sales = SUM(DatacoSupplyChainDataset[Sales])
Total Profit = SUM(DatacoSupplyChainDataset[Order Profit Per Order])
Total Orders = DISTINCTCOUNT(DatacoSupplyChainDataset[Order Id])
```

Year-to-Date (YTD)

Accumulates the measure from the start of the calendar year to the current date in context.

```
Sales YTD = TOTALYTD([Total Sales], DateTable[Date])
Profit YTD = TOTALYTD([Total Profit], DateTable[Date])
```

Quarter-to-Date (QTD)

```
Sales QTD = TOTALQTD([Total Sales], DateTable[Date])
```

Year-over-Year (YoY)

Previous year comparison and growth percentage:

```
Sales PY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date]))
YoY Growth % = DIVIDE([Total Sales] - [Sales PY], [Sales PY])
```

A separate card measure was also created because card visuals do not automatically inherit year context from slicers:

```
YoY Growth Card =
VAR CurrentYear = MAX(DateTable[Year])
VAR CurrentSales = CALCULATE([Total Sales], DateTable[Year] = CurrentYear)
VAR PreviousSales = CALCULATE([Total Sales], DateTable[Year] = CurrentYear - 1)
RETURN DIVIDE(CurrentSales - PreviousSales, PreviousSales)
```

Visuals Implemented

- KPI cards for Total Sales, Total Profit, Total Orders and YoY Growth %
- Line chart showing Sales YTD trend by Month
- Clustered column chart comparing Total Sales vs. Sales Previous Year by Year

Note: The negative YoY result in 2018 is due to fewer transaction records in that year - a data completeness issue, not a business decline.

3. Rolling Window Analytics – 30 & 90-Day Averages

Contributor: Sairam

Rolling window measures smooth out short-term noise in sales data by averaging values over a fixed trailing window. This reveals the true underlying momentum in supply chain performance, making trend changes easier to identify.

Rolling 30-Day Sales

Total sales over the trailing 30 days ending on the current date in context:

```
Rolling 30D Sales =  
CALCULATE(  
    [Total Sales],  
    DATESINPERIOD(DimDate[Date], MAX(DimDate[Date]), -30, DAY)  
)
```

Rolling 30-Day Average

```
Rolling 30D Avg = DIVIDE([Rolling 30D Sales], 30)
```

Rolling 90-Day Average

The 90-day average gives a longer-horizon baseline that absorbs seasonal fluctuations:

```
Rolling 90D Avg =  
DIVIDE(  
    CALCULATE(  
        [Total Sales],  
        DATESINPERIOD(DimDate[Date], MAX(DimDate[Date]), -90, DAY)  
    ),  
    90  
)
```

Chart Observations

A dual-line chart was constructed where each measure was plotted against Jan 2015 to Jan 2018, and a trend line overlaid.

The important findings of the visual:

- The highest sales were at \$400 in and around Jan 2016 - the most robust point in the entire period.
- A noteworthy trough of 130 struck mid-2016, an indicator that demand was declining or there was a supply shock.
- Recovery to \$300 in early 2017 was succeeded by a prolonged downward trend to the end of the year.
- The Rolling 90-Day Avg uses the 30-Day line but flattens the steeper inclines and declivities.
- The general dashed trend line presents a decreasing trend between the period of about 300 to about 180.

4. Risk Scoring Model & Methodology

Contributor: Karishma | Documented by: Khushi Sahni

This section is the formal Risk Methodology Document deliverable for Week 3.

4.1 Objective

The Risk Scoring Model is a supply chain risk assessment model that is based on the evaluation of three main indicators of operational and financial stress at an individual order level. Every order is given a complex numerical risk rating and automatically categorized into a risk level - Low, Medium, or High. This will enable the supply chain managers to prioritize on the intervention, concentrate on the most troubled shipments, and track the risk trends through time.

4.2 Risk Factors Selected

Risk Factor	Why It Was Selected	Data Field
Late Delivery	Delivery failures directly damage customer satisfaction and increase handling costs	Late_delivery_risk, Days for shipping (real/scheduled)
Profit per Order	Negative or very low profit indicates unsustainable order economics - often driven by over-discounting or high fulfilment cost	Order Profit Per Order
Discount Rate	Excessive discounting erodes margins and can signal demand weakness or pricing misalignment	Order Item Discount Rate

4.3 Scoring Methodology

All the factors are then translated to the 0-100 scale independently with the SWITCH(TRUE()) pattern of Power BI. This normalisation makes all the three factors dimensionally comparable to ensure that they are combined. The score of 100 is maximum risk; score of 10 is minimum risk.

Delay Score

```
Delay Score =  
SWITCH(  
    TRUE(),  
    'Fact_Orders'[Late_delivery_risk] = 1,  
100,  
    'Fact_Orders'[Days for shipping (real)] -  
        'Fact_Orders'[Days for shipment (scheduled)] > 3,  
70,  
    'Fact_Orders'[Days for shipping (real)] >  
        'Fact_Orders'[Days for shipment (scheduled)],  
40,  
    10  
)
```

Profit Score

```
Profit Score =  
SWITCH(  
    TRUE(),  
    'Fact_Orders'[Order Profit Per Order] < 0,    100,  
    'Fact_Orders'[Order Profit Per Order] < 50,    70,  
    'Fact_Orders'[Order Profit Per Order] < 100,    40,  
    10  
)
```

Discount Score

```
Discount Score =  
SWITCH(  
    TRUE(),  
    'Fact_Orders'[Order Item Discount Rate] > 0.4,    100,  
    'Fact_Orders'[Order Item Discount Rate] > 0.2,    70,  
    'Fact_Orders'[Order Item Discount Rate] > 0.1,    40,  
    10  
)
```

4.4 Scoring Reference Table

Factor	Condition	Score
Delay	Late delivery flag = 1 (confirmed late)	100
Delay	Real shipping days exceed scheduled by more than 3	70
Delay	Real shipping days exceed scheduled by any amount	40
Delay	Delivered on time	10
Profit	Order profit is negative (loss-making)	100
Profit	Profit is between \$0 and \$50	70
Profit	Profit is between \$50 and \$100	40
Profit	Profit exceeds \$100	10
Discount	Discount rate above 40%	100
Discount	Discount rate between 20% and 40%	70
Discount	Discount rate between 10% and 20%	40
Discount	Discount rate at or below 10%	10

4.5 Composite Risk Score

Weighted averaging is used to combine the three factor scores into one composite score. Weights indicate the relative operational significance of each factor:

```
Risk Score =  
(  
    'Fact_Orders'[Delay Score] * 0.45 +  
    'Fact_Orders'[Profit Score] * 0.30 +  
    'Fact_Orders'[Discount Score] * 0.25  
)
```

- Delay weightage = 45% - shipping failure is the most immediate and quantifiable risk to the customer.
- Profit weightage = 30% - unprofitable orders are a fundamental financial sustainability issue.
- Discount weightage = 25% - a leading indicator of margin pressure, partially associated with profit.

4.6 Risk Classification

The composite score is classified into three tiers:

```
Risk Level =  
SWITCH(  
    TRUE(),  
    'Fact_Orders'[Risk Score] >= 70, "High",  
    'Fact_Orders'[Risk Score] >= 40, "Medium",  
    "Low"  
)
```

Risk Level	Score Range	Recommended Action
High	70 and above	Immediate review - escalate to operations team
Medium	40 to 69	Monitor closely - flag for weekly review cycle
Low	Below 40	Routine tracking - no immediate action required

4.7 Supporting Measures

```
Avg Risk Score =  
AVERAGE('Fact_Orders'[Risk Score])  
  
High Risk % =  
DIVIDE(  
    CALCULATE(  
        COUNT('Fact_Orders'[Order Id]),  
        'Fact_Orders'[Risk Level] = "High"  
    ),  
    COUNT('Fact_Orders'[Order Id])  
)
```

4.8 Model Results

Metric	Value
Average Risk Score	51.01 out of 100
High Risk Orders	25.03%
Medium Risk Orders	37.73%
Low Risk Orders	37.24%
Total Orders Analysed	180,519

4.9 Key Insights

- High Risk is approximately 1 per 4 orders - a high percentage that is worth constant attention.
- Delivery delay is the predominant factor. The Standard shipping (44.77) is worse in terms of average delay score than the First Class shipping (57.18) even though the former costs less than the latter, which is counter-intuitive.
- High Risk: The model predicts the logic of High Risk, which is almost entirely the case with the loss-making orders.
- High discount rates compound risk - they both depreciate margins and are commonly accompanied by fulfilment delays.
- High-Risk orders are concentrated the most in Africa and LATAM markets, which indicates that there are regional fulfilment issues to focus on.

5. Customer & Product Segmentation

Contributor: Arun Dande

Segmentation measures classify customers and products into strategic tiers, enabling targeted analysis of high-value relationships and top-performing product lines.

Customer Segmentation

```
Customer Sales =  
CALCULATE(  
    [Total Sales],  
    ALLEXCEPT(DimCustomer, DimCustomer[Customer Id])  
)  
  
Customer Tier =  
IF([Customer Sales] > 10000, "High Value", "Standard")
```

The ALLEXCEPT pattern ensures each customer's total lifetime sales are evaluated independently of any active report filters - giving accurate tier assignments regardless of the current visual context.

Product Segmentation


```
Product Segment =  
IF([Total Sales] > 5000, "Top Product", "Regular")
```

This enables a Pareto-style view - identifying the smaller set of products that contribute disproportionately to total revenue.

6. Drillthrough Pages

Contributor: Aditya

Drillthrough pages allow contextual navigation into the report - a right-clicking on a piece of data in any summary visual takes the user to a special detail page that is scoped to the context. This makes the report a self-service investigation tool and less external data exporting is required.

How Drillthrough Works in Power BI

- A special report page is built and set up as a drillthrough destination.
- One or more fields (e.g. Product Category, Region, Customer) are entered in the Drillthrough filter on that page.
- All the images on the drillthrough page will automatically filter to the chosen value on the source page.
- Power BI will automatically add a back-navigation button to allow users to go back to the summary view.

Implementation

The drillthrough page was constructed by Aditya on the primary analytics PBIX file. The drillthrough will be set to take a category or region selection on the summary visuals so that its users can navigate between high-level metrics to order-level detail in a single interaction - without leaving the report.

7. Bookmarks & Dynamic Measures

Contributor: Shreeram Prajapati

Bookmarks

Bookmarks record and preserve a particular report state - state of the filters in use, state of the slicers in use, state of the visuals in view. They enable the users to toggle between pre-configured views without necessarily having to filter them manually.

Five bookmarks were created:

Bookmark Name	What It Captures
Report_2015-16	Year slicer filtered to 2015–2016; shows early-period performance
Report_2015-17	Year slicer extended to 2015–2017; useful for mid-period trend comparison
Report_2015-18	Full date range 2015–2018; the default report state
TotalSales_Southern Europe	All visuals filtered to the Southern Europe order region

TotalSales_Central Asia	All visuals filtered to the Central Asia order region
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Dynamic Measures

A measure selection was developed dynamically such that one visual can show multiple measures based on a slicer selection - without visuals or pages being duplicated. This makes the report clean and still provides complete metric flexibility.

Step 1 – Create the Measure Selector Table

```
Measure Selector =
DATATABLE(
    "MeasureName", STRING,
    {
        { "Sales" },
        { "Profit" },
        { "Quantity" },
        { "Rolling 30-Day Avg" },
        { "Rolling 90-Day Avg" }
    }
)
```

Step 2 – Add as a Slicer

MeasureName column is added to a slicer visual in a single-select mode, and therefore only one metric is enabled.

Step 3 – Create the Dynamic Measure

```
Dynamic Measure =
SWITCH(
    SELECTEDVALUE(MeasureSelector[MeasureName]),
    "Sales", SUM(FactOrders[Sales]),
    "Profit", SUM(FactOrders[Order Profit Per Order]),
    "Quantity", SUM(FactOrders[Order Item Quantity]),
    "Rolling 30-Day Avg", [Rolling 30D Avg],
    "Rolling 90-Day Avg", [Rolling 90D Avg]
)
```

As a user picks on any metric in the slicer, all visuals that are connected to the Dynamic Measure are updated immediately. The before/after effect is evident in the documentation of Shreeram - the visual table in the form of a table is empty and then it fills with Profit, Quantity, Rolling averages, or Sales based on the choice.

8. Master DAX Measures Reference

Full catalogue of all measures implemented across the project, sourced from Dhanlaxmi's DAX Documentation file:

Measure	Formula	Purpose & Report Page
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Total Sales	SUM(DatacoSupplyChainDataset[Sales])	Base revenue measure - used across all pages
Total Profit	SUM(DatacoSupplyChainDataset[Order Profit Per Order])	Base profit - Time Intelligence Dashboard
Total Orders	DISTINCTCOUNT(DatacoSupplyChainDataset[Order Id])	Order count - Time Intelligence Dashboard
Sales YTD	TOTALYTD([Total Sales], DateTable[Date])	Cumulative YTD sales - Time Intelligence
Profit YTD	TOTALYTD([Total Profit], DateTable[Date])	Cumulative YTD profit - Time Intelligence
Sales QTD	TOTALQTD([Total Sales], DateTable[Date])	Quarter-to-date sales - Time Intelligence
Sales PY	CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date]))	Previous year sales - Time Intelligence
YoY Growth %	DIVIDE([Total Sales]-[Sales PY],[Sales PY])	Year-over-year growth rate - Time Intelligence
Rolling 30D Sales	CALCULATE([Total Sales], DATESINPERIOD(...,-30,DAY))	Trailing 30-day total - Trend Analysis
Rolling 30D Avg	DIVIDE([Rolling 30D Sales], 30)	Daily average over 30 days - Trend Analysis
Rolling 90D Avg	DIVIDE(CALCULATE(...,-90,DAY), 90)	Daily average over 90 days - Trend Analysis
Delay Score	SWITCH(TRUE(), Late_delivery_risk=1,100,...,10)	Order-level delay risk - Risk Dashboard
Profit Score	SWITCH(TRUE(), Profit<0,100,...,10)	Order-level profit risk - Risk Dashboard
Discount Score	SWITCH(TRUE(), Discount>0.4,100,...,10)	Order-level discount risk - Risk Dashboard
Risk Score	[Delay]*0.45 + [Profit]*0.30 + [Discount]*0.25	Composite risk score (0–100) - Risk Dashboard
Risk Level	SWITCH(TRUE(), Score>=70,"High", Score>=40,"Medium","Low")	Risk tier - Risk Dashboard
Avg Risk Score	AVERAGE('Fact_Orders'[Risk Score])	Average risk across all orders - Risk Dashboard

High Risk %	DIVIDE(CALCULATE(COUNT(OrderId), RiskLevel='High'), COUNT(OrderId))	Proportion of high-risk orders - Risk Dashboard
Customer Sales	CALCULATE([Total Sales], ALLEXCEPT(DimCustomer, Customer Id))	Per-customer total sales - Segmentation
Customer Tier	IF([Customer Sales]>10000,'High Value','Standard')	Customer value tier - Segmentation
Product Segment	IF([Total Sales]>5000,'Top Product','Regular')	Product performance tier - Segmentation
Dynamic Measure	SWITCH(SELECTEDVALUE(MeasureSelector[MeasureName]),...)	Slicer-driven metric - Dynamic Dashboard

9. Key Analytical Insights

Synthesised findings from all analytical modules:

Time Intelligence

- YTD: Good Q1 accumulation in most years is confirmed, as per the patterns of post-holiday supply chain restocking.
- In the 2018 negative YoY, this is a data completeness artifact - there are fewer records of transactions and not a decrease in revenue.

Rolling Window Analytics

- The most unusual point of the dataset is the 2016 trough (\$130) which has to be drilled down by product category and region.
- 90-Day rolling average: The structural negative trend in the 90-Day rolling average is evident since the middle of the year 2016 and is not dependent on monthly noise.

Risk Scoring

- High Risk 25 percent - systematic actions around shipping performance and discount policies have the potential to have a material impact on this.
- The delay paradox of First-Class shipping (the highest delay score, yet the highest-level provider) indicates the possibility of bottlenecks in capacity or processing operations that should be operationally researched.
- The profit-risk relationship confirms the model - loss making orders are concentrated almost in the High-Risk tier.

Segmentation

- High-Value customers: This is a small segment of the total customers that generate a disproportional amount of revenue - classic Pareto distribution.

- Top Products demonstrate fewer average risk scores, indicating that there are more predictable and reliable fulfilment profiles of high-volume product lines.

Drillthrough & Bookmarks

- Drillthrough pages Drillthrough pages allow the report to be converted to a self-service exploration tool - a single click can take you to order detail by region or to individual order detail.
- Bookmarks can cut analysis time drastically when you have to revisit the same analysis again and again - it only takes a single click to change between the 2015-16, 2015-17, and 2015-18 views, which would otherwise require you to manually adjust the filters.

10. Conclusion

The third week is the most technical week of the programme to date. The team has developed a fully operational production quality Power BI solution across seven analytical modules that is far more than mere, static dashboards - the solution facilitates dynamic metric selection, interactive navigation, automatic classification of risks, and contextual drill-through investigation.

The Risk Scoring Model is the main feature of the analytical work of the week. It provides an immediate action signal to supply chain managers because it is a single weighted composite of three separately rated factors. The conclusion that 25% of orders are High Risk - due to the fact that the business is mostly driven by delivery delays - gives the business clear operational priorities.

The final step in integration is the complete PBIX consolidation since all the modules are now submitted. The base established this week gives a base to the subsequent improvements - more risk factors, predictive scoring, or region-specific threshold calibration - in subsequent iterations.

Tools and technologies used across the project:

- Microsoft Power BI Desktop - Report development, data modelling, and visualisation
- DAX (Data Analysis Expressions) - All calculated measures and calculated columns
- Power Query (M) - Data table generation and data transformation
- DataCo Supply Chain Dataset - Primary data source
- Microsoft Excel - DAX measures documentation catalogue
- Microsoft Word - Narrative documentation